Renal Dysfunction / Renal Disorders

Evaluation

Chronic kidney disease is diagnosed when there is evidence of kidney damage for at least 3 months or in any patient with a GFR of less than 60 mL/min for that same amount of time.[50][51] To calculate estimated GFR (eGFR), 3 equations have been commonly used (the Modification of Diet in Renal Disease Study, CKD-Epidemiology Collaboration [CKD-EPI], and Cockcroft-Gault formula). Until recently, the CKD-EPI equation was commonly used to calculate eGFR from serum creatinine, age, and race in the majority of health systems. In 2021, a joint American Society of Nephrology(ASN) and the National Kidney Foundation (NKF) task force recommended using a new CKD-EPI 2021 equation that does not include the race coefficient, which is now the current standard of care.[52]

However, the equation tends to underestimate actual GFR when the GFR is greater than 60 mL/min/1.73 m².[53] The ASN/NKF task force also recommended the use of cystatin C in calculating eGFR. Cystatin C is a protein secreted by all nucleated cells, is freely filtered by the kidneys, and has a near-complete absorption; hence, its levels are less affected by muscle mass, diet, or nutritional status.[54] The newer eGFR equation, which incorporates serum creatinine and cystatin C, is considered more accurate and has fewer differences between Black and non-Black populations.[55] Further evaluation of kidney disease can include a renal ultrasound, complete blood count, basic metabolic panel, urinalysis, and/or kidney biopsy.

Complete Blood Count

Anemia is common in patients with ESRD, and it is unusual for these patients to have normal hemoglobin levels. Anemia is associated with decreased quality of life and increased mortality. The landmark Normal Hematocrit Trial, along with several other trials, actually showed higher rates of heart failure and thrombotic complications when hemoglobin was corrected to greater than 11.0 mg/dL.[56][57]

Basic Metabolic Panel

The blood urea nitrogen and serum creatinine levels are elevated. Hyperkalemia and low bicarbonate levels are usually present before dialysis (whose goal is to normalize electrolytes). Serum albumin levels are low due to urinary protein loss or malnutrition. Serum phosphate, 25-hydroxyvitamin D, alkaline phosphatase, and intact PTH levels are obtained to look for evidence of renal bone disease.[58] A lipid profile should be obtained because of the risk of cardiovascular disease.

Urinalysis

A spot urine protein/creatinine ratio can be used to quantify albuminuria. A value higher than 30 mg of albumin per gram of creatinine is considered 'moderately increased albuminuria', while values greater than 300 mg/g are considered 'severely increased albuminuria'.

Additionally, a 24-hour urine protein test can also be performed; a value greater than 3.5 g is concerning for nephrotic-range proteinuria.

Renal Ultrasonography

Renal ultrasonography should be performed to look for hydronephrosis or involvement of the retroperitoneum with fibrosis, tumor, or diffuse adenopathy. Small, echogenic kidneys are observed in advanced renal failure. Structural abnormalities, such as polycystic kidneys, may also be observed on ultrasonograms. An ultrasound can provide data estimating size, obstructions, stones, echogenicity, and cortical thinning.[59]

Radiology

Plain abdominal radiography can detect radio-opaque stones or nephrocalcinosis, while a voiding cystourethrogram is diagnostic for vesicoureteral reflux.[60] Computed tomography scanning can help better describe renal masses and cysts and is also sensitive for identifying renal stones. Magnetic resonance angiography can accurately diagnose renal artery stenosis. A renal radionuclide scan with captopril administration can help diagnose renal artery stenosis and quantify the differential renal contribution to the total GFR.

Renal Biopsy

Percutaneous ultrasound-guided renal biopsy is indicated when the diagnosis is unclear after an appropriate workup.[61]

Specific Tests

- Serum and urine protein electrophoresis for multiple myeloma
- Antinuclear antibodies (ANA), double-stranded deoxy ribonucleic acid antibody levels for systemic lupus erythematosus
- Serum complement levels
- Cytoplasmic and perinuclear pattern antineutrophil cytoplasmic antibody levels for granulomatosis with polyangiitis (Wegener granulomatosis) and microscopic polyangiitis
- Anti–glomerular basement membrane antibodies for Goodpasture syndrome
- Hepatitis B and C, human immunodeficiency virus, and venereal disease research laboratory serology

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Treatment / Management

Treatment of CKD or ESRD involves correcting parameters at the level of the patient's presentation.[62] Interventions aimed at slowing the rate of kidney disease should be initiated and can include:

- The baseline treatment involves addressing the underlying cause and managing blood pressure and proteinuria. Blood pressure should be targeted to a systolic blood pressure of less than 130 mm Hg and a diastolic blood pressure of less than 80 mm Hg in adults with or without diabetes mellitus whose urine albumin excretion exceeds 30 mg for 24 hours. For diabetic individuals with proteinuria, an ACEI or ARB should be started in cases where urine albumin values range between 30 and 300 mg in 24 hours and greater than 300 mg in 24 hours. These drugs slow disease progression, particularly when initiated before the GFR decreases to less than 60 mL/min or before the plasma creatinine concentration exceeds 1.2 mg/dL in women and 1.5 mg/dL in men.[63]
- Other targets in preventive care and monitoring should include achieving tight glycemic control, reducing cardiovascular risk, controlling hypertension, and implementing general lifestyle recommendations such as smoking cessation and dietary restriction of salt. Glycemic control and lipid control are critical. A glycated hemoglobin A1C of less than 7% is generally recommended to prevent or delay microvascular complications in this population. Management with sodium-glucose transporter 2 inhibitors may reduce the disease burden in those with type 2 diabetes mellitus.[64]

Treatment for Specific Indications

Hyperkalemia

Many advances have been made regarding the treatment of hyperkalemia. In general, avoiding triggers for hyperkalemia is recommended. Reducing potassium-containing foods has long been recommended; however, recent evidence suggests that this approach may have limited effects and could lead to nutrient deficiencies in already malnourished patients.[31][65] Many patients with CKD/ESRD have concomitant heart disease requiring potassium-raising drugs such as ACEIs, ARBs, neprilysin inhibitors, mineralocorticoid antagonists, and other necessary drugs that cannot be discontinued. Notably, treating metabolic acidosis is also an essential component of managing hyperkalemia, as potassium shifts from the intracellular to the extracellular space. For short-term, immediate management of hyperkalemia, please see StatPearls' companion reference, "Hyperkalemia."

The long-term treatments of choice for hyperkalemia include the following:

- Sodium polystyrene sulfonate has traditionally been used; however, this is often poorly tolerated by patients and has many gastrointestinal (GI) adverse events. Rarely has it been associated with serious or fatal adverse events such as colonic necrosis.[31][65][66]
- Newer potassium binders have started to replace the above. These drugs exchange potassium for other drugs in the GI tract and tend to be better tolerated than sodium polystyrene sulfonate. Sodium zirconium cyclosilicate exchanges potassium for sodium and hydrogen ions. The most common adverse event is edema.

- Patiromer acts similarly except that it exchanges potassium for calcium ions. The most common adverse events are GI: constipation, diarrhea, nausea, abdominal discomfort, and flatulence.[31][65][67]

Metabolic acidosis

Protein metabolism creates an acid load, while fruits and vegetables contain alkaline compounds, which neutralize the acid. Decreased ammoniagenesis is associated with reduced glutamine uptake in the proximal tubule, particularly in patients with high protein intake and low consumption of fruits and vegetables.[36] Several studies have shown a slowing of GFR decline with the use of bicarbonate, citrate, and a diet rich in fruits and vegetables. Therefore, all patients with a sodium bicarbonate level of less than 22 mEq/L be treated for acidemia.[68][69][70] One study found that treatment of acidemia with fruits and vegetables for 1 year improved parameters without causing hyperkalemia.[70][71]

Fluid titration

Fluid status is closely monitored in all patients on renal replacement therapy. This is achieved through regular weight checks and clinical assessments. Patients on peritoneal dialysis and home hemodialysis are especially encouraged to check daily weights. If a patient on hemodialysis is very fluid-overloaded, ultrafiltration can be done for fluid removal alone.

Anemia

As noted above, management of anemia in patients with ESRD is very complex. Please refer to StatPearls' comprehensive review of "Anemia of Chronic Disease," for further information on this complicated topic. Briefly, as per KDIGO Guidelines, erythropoiesis-stimulating agents (ESAs) are typically considered for patients with CKD when hemoglobin levels drop below 10 g/dL. However, ESA treatment is individualized based on factors such as anemia symptoms, transfusion requirements, the rate of hemoglobin decline, and response to iron therapy. Erythropoietin (50-100 units/kg) is usually administered intravenously or subcutaneously every 1 to 2 weeks, while darbepoetin alfa is dosed every 2 to 4 weeks. For patients on dialysis, erythropoietin is administered with each dialysis session (3 times a week), whereas darbepoetin alfa is administered once a week. The biosimilar epoetin alfa-epbx was also approved for use in the US in 2018. Continuous erythropoiesis receptor activator is a newer, longer-acting ESA that may be preferred over other ESAs due to its lower administration frequency.

A crucial treatment component is that patients must be iron replete for the above medications to be effective. KDIGO recommends a target transferrin saturation between 20% and 30% and ferritin levels between 100 to 500 ng/mL in patients with CKD and anemia. The European Renal Best Practice Guidelines (2013) propose a ceiling for TSAT at 30% and ferritin at 500 ng/mL. Additionally, dialysis centers often have their own specific goals and protocols.[72] Guidelines from the National Institute for Healthcare and Excellence (2015) and the Renal Association (2017) suggest using a ferritin ceiling of 800 ng/mL.[73]

Bone disease

As noted above, please see StatPearls' comprehensive review, "Chronic Kidney Disease-Metabolic bone disorder," for a review of this very complex topic.

Patients on dialysis have the following recommended targets of therapy:

- Phosphate levels are typically targeted to be between 3.5 and 5.5 mg/dL (1.13-1.78 mmol/L) in patients on dialysis.
- Serum calcium levels are ideally maintained below 9.5 mg/dL (less than 2.37 mmol/L); cincalcet is often required for this.
- PTH levels should be maintained at less than 2 to 9 times the upper limit for the assay.[74]
- Vitamin D is often replaced with calcitriol or vitamin D analogs.

Once hyperphosphatemia is under control, PTH management is based on trends rather than isolated laboratory values. Notably, it is not advisable to suppress PTH to less than twice the upper limit, as this may lead to adynamic bone disease.[75] Phosphate binders used include calcium phosphate, calcium acetate, sevelamer, and lanthanum.

Planning for Long-term Renal Replacement Therapy

Early patient education should be initiated regarding the natural progression of the disease, various modalities for dialysis, and renal transplantation. For patients in whom transplantation is not imminent, a primary arteriovenous fistula should be created in advance of the anticipated date of dialysis.[76] The optimal timing of dialysis initiation (early vs late) in patients with CKD is unknown. [77] Every patient with end-stage renal disease should be timely referred for renal transplantation. Referral for kidney transplantation evaluation is generally done when the eGFR falls to less than 20 ml/min/1.73m^2.[78] As noted above, early planning leads to improved outcomes. Patients should be educated about the options of in-center hemodialysis, peritoneal dialysis, home hemodialysis, and preemptive kidney transplant.

Over 80% of patients in the US are initiated on dialysis via a central venous catheter, which is associated with significantly higher infection rates than permanent access. Permanent access requires preparations (usually once GFR is less than 20 mL/min/1.73m^2), as the mean time for arteriovenous fistula maturation is 3 months, and many fistulas do not mature for dialysis use initially. Arteriovenous grafts can usually be used sooner than fistulas, at around 2 to 3 weeks post-placement; however, they usually last only 2 to 3 years and require more frequent interventions to maintain patency.[5] All potential modalities should be discussed with patients in an interdisciplinary team setting, and patients should be closely monitored once they reach stage 4 CKD.

Chronic Pancreatitis

Evaluation

When chronic pancreatitis is suspected, the new mechanistic definition can aid in diagnosis. This approach integrates clinical features (typical signs, symptoms, and family history) with risk factors (see **Table.** Modified TIGAR-O Pancreatitis Risk/Etiology Checklist). Biomarkers further support the diagnosis and may include the following: [23]

- Serum markers
    - Amylase
    - Lipase
    - Glucose
    - IgG4
    - Triglycerides
    - Tumor markers
- Nutritional deficiencies
    - Hypovitaminosis A, D, K, and B12
- Additional tests
    - Sweat chloride levels
    - Pancreatic biopsies

The 72-hour quantitative fecal fat collection is the most definitive test for advanced pancreatic insufficiency to evaluate steatorrhea. A fecal fat excretion of greater than 7 g/d confirms the diagnosis.

Alternatively, a fecal elastase-1 level can be performed using a single random stool sample to assess pancreatic insufficiency.[24]

- Normal fecal elastase level: > 500 µg/g of stool
- Borderline range: 200 to 500 µg/g (nonspecific, may be normal)
- Moderate exocrine pancreatic insufficiency: 100 to 200 µg/g
- Severe exocrine pancreatic insufficiency: <100 µg/g

Although neither test is highly accurate, fecal elastase testing is less reliable than the 72-hour fecal fat collection but is more practical and widely used in clinical practice. For patients with a history of acute pancreatitis, chronic pancreatitis-related pain, and signs of maldigestion, such as steatorrhea or fat-soluble vitamin deficiencies, confirmatory imaging with CT or magnetic resonance cholangiopancreatography testing (MRCP) should be the initial diagnostic step.

Diagnostic Algorithm

- MRCP and CT are the preferred imaging modalities.
  - Detect calcifications (hallmark sign), pancreatic enlargement, and ductal obstruction or dilation.
  - MRCP is more sensitive and specific for chronic pancreatitis than transabdominal ultrasound or plain x-rays, though both can reveal calcifications.
- Endoscopic ultrasound (EUS) is an alternative, particularly for early chronic pancreatitis before fibrosis develops.
  - More invasive and less specific than MRCP or CT.
  - Reserved for cases where MRCP or CT is nondiagnostic but clinical suspicion remains high.[23][25][26]
- Secretin-enhanced MRCP may be used if standard imaging and EUS are inconclusive.
- Endoscopic retrograde cholangiopancreatography (ERCP) was the previous gold standard but is now largely supplanted by MRCP.
  - This technique is still used if therapeutic intervention, such as biliary or pancreatic sphincterotomy, stent placement, or stone extraction, is needed.
- Pancreatic biopsy is indicated if clinical suspicion is high, but imaging remains nondiagnostic.
- Direct and indirect exocrine function tests can be performed at any point in the diagnostic process but should complement imaging studies rather than replace them.[24]
  - Direct pancreatic function tests:
    - The cholecystokinin (CCK) stimulation test measures trypsin or lipase and is sensitive to early exocrine pancreatic insufficiency. However, it is limited in availability and uncomfortable for patients.
    - Endoscopic secretin stimulation test measures bicarbonate secretion but requires upper gastrointestinal endoscopy, making it costly and invasive.
  - Indirect pancreatic function tests:
    - The ^{13}C-mixed triglyceride test is 90% sensitive but takes up to 6 hours to perform and is not widely available.
  - The fecal elastase test is widely available and practical.
  - Serum testing:
    - Trypsin or trypsinogen levels are no longer recommended, as they lack specificity and correlation with imaging findings.

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The primary goals of treatment are managing pain and improving malabsorption. Abdominal pain in chronic pancreatitis arises from inflammation, neuropathic mechanisms, and ductal obstruction. Once established, chronic pain is typically persistent and lifelong without intervention and can be debilitating.[27]

One of the most critical steps in management is encouraging alcohol cessation, as continued alcohol use can exacerbate pain and accelerate disease progression. Similarly, smoking cessation is vital and essential, as smoking is known to worsen the condition over time.

Intervention is not clearly indicated when ductal strictures or stones are present, but the patient remains asymptomatic. However, if these obstructions cause symptoms, therapeutic options such as endoscopic procedures, surgical intervention, or extracorporeal shock wave lithotripsy (ESWL) may be considered.

Pain Management of Chronic Pancreatitis

Pain management in chronic pancreatitis involves medical, endoscopic, percutaneous, and surgical interventions. When imaging reveals ductal obstruction due to strictures or stones, the initial treatment approach should include endoscopy therapy or, in the case of stones, ESWL. Surgical intervention may be necessary if these interventions are unsuccessful or technically unfeasible. However, if no obstruction is identified, a trial of medical therapy is the recommended first-line approach.[28]

Medical management: The initial approach to pain management in chronic pancreatitis should prioritize non-opioid medications before considering opioids. First-line options include acetaminophen or paracetamol, nonsteroidal anti-inflammatory drugs, pregabalin, selective serotonin reuptake inhibitors, and tricyclic antidepressants.

Opioids should be avoided whenever possible due to the risks of dependence or addiction, tolerance, and adverse effects. However, they may be considered when non-opioid treatments prove ineffective or inadequate. Opioids should be prescribed in combination with non-opioid agents, particularly gabapentin, and initiated with a milder opioid such as tramadol, which has fewer gastrointestinal adverse effects and may be more effective when combined with pregabalin.

Antioxidant combinations—including selenium, ascorbic acid, β-carotene, L-methionine, and sometimes α-tocopherol—have demonstrated some effectiveness in pain relief in early chronic pancreatitis based on small studies. However, variability in formulations and dosages across studies limits definitive conclusions. Despite this, antioxidant therapy is safe and remains an option for some patients.[29][30]

Pancreatic enzyme supplementation, although expensive, has not been proven effective in relieving pain associated with chronic pancreatitis.[31] However, it may still be beneficial for patients with exocrine pancreatic insufficiency to improve nutrient absorption and digestion

Endoscopic management: A celiac plexus block, involving the injection of a local anesthetic (typically bupivacaine), sometimes combined with a corticosteroid (typically triamcinolone), can be performed endoscopically using EUS or with CT guidance by a radiologist.[32] Pain relief from this procedure typically lasts 3 to 6 months and can be repeated as needed.

Obstructing pancreatic stones can be extracted using a basket, whereas balloons are rarely effective except in children, who may have softer stones. If traditional extraction methods fail, ESWL, pancreatoscopy-assisted laser, or electrohydraulic lithotripsy may be considered, particularly when a guidewire cannot be advanced past the stone(s).

For main pancreatic duct dilation, ERCP can help decompress ductal obstruction through pancreatic sphincterotomy, stone extraction, stricture dilatation, or stent placement, depending on diagnostic pancreatographic findings. However, when chronic pancreatitis is advanced enough to cause steatorrhea, ductal decompression via dilatation, stenting, or stone extraction offers limited benefit. During ERCP, ductal cytologic brushing can also be performed to evaluate for malignancy.

Pancreatic duct leaks may be treated with the temporary placement of a plastic stent to facilitate healing. Endoscopic drainage into the gastrointestinal tract, typically the stomach, can be performed using 1 or more lumen-apposing metal stents for peri-gastric pancreatic fluid collections such as pseudocysts or necrosis.[33]

Surgical management: Surgical decompression is preferred over ERCP when the main pancreatic duct is narrowed. Several surgical procedures are available based on the disease's location and severity:

- Puestow procedure (most commonly performed): Lateral pancreaticojejunostomy combined with a Roux-en-Y pancreaticojejunostomy to improve drainage.
- Beger procedure: Duodenal-preserving pancreatic head resection with a Roux-en-Y pancreaticojejunostomy, used for pancreatic head-dominant chronic pancreatitis.
- Frey procedure: A modification of the Puestow procedure that involves lateral pancreaticojejunostomy draining the pancreatic head and uncinate process. Unlike the Beger procedure, it preserves the pancreatic neck while removing part of the overlying pancreatic head.
- Distal pancreatectomy: Typically performed laparoscopically for tail-predominant disease. However, complications such as pancreatic fistulae and endocrine failure are relatively common.[34]
- Total pancreatectomy with islet autotransplant: Reserved for patients with refractory pain who have failed all other treatments. Although once used selectively, its application has been increasing in recent years.[35]

Additional surgical indications for chronic pancreatitis include:

- Pancreatic abscess, fistula, or pseudocyst when endoscopic management fails
- Pancreatic ascites
- Stenosis of the duodenum leading to gastric outlet obstruction

- Gastric variceal bleeding due to splenic vein thrombosis when conservative therapy is ineffective [36]

In severe cases, inpatient care may be necessary, particularly for patients with chronic pain and anorexia, often requiring narcotics and nutritional support. Pancreatic enzyme supplementation, typically taken with meals, may help reduce pain, although its effectiveness remains uncertain.

Management of Exocrine Pancreatic Insufficiency

Chronic pancreatitis accompanied by weight loss, malnutrition, diarrhea, steatorrhea, osteopenia, or osteoporosis should raise suspicion of exocrine pancreatic insufficiency. Pancreatic enzyme replacement therapy is recommended for managing symptomatic exocrine pancreatic insufficiency. The potential benefits of pancreatic enzyme replacement therapy include improved symptoms, reduced morbidity related to maldigestion (especially bone fractures), better absorption of fat-soluble vitamins and trace elements, reduced fecal fat, weight gain, improved quality of life, and potentially lower mortality rates.[1]

A reduced fecal elastase level is the accessible diagnostic test for moderate-to-severe exocrine pancreatic insufficiency. Patients should avoid consuming a high-fiber diet (>25 g/d), as excessive fiber may interfere with enzyme activity. However, a low-fat diet is no longer recommended, as it may contribute to nutritional deficiencies.

Pancreatic enzyme replacement therapy should be taken at the start of each meal or snack, with additional supplementation added during or after the meal if needed (see **Table 3.** Dosage of Pancreatic Enzyme Replacement Therapy). Failure to respond to maximal doses of PERT raises several considerations. To maintain effectiveness, capsules must be appropriately stored at temperatures below 25 °C or 77 °F and kept away from direct sunlight, cars, or heat sources. In addition, inadequate response may indicate the need for acid suppression therapy with a histamine-2 receptor antagonist or proton pump inhibitor. If symptoms persist despite these measures, an alternative cause should be considered.[37]

Cholestatic Jaundice

Evaluation

Diagnostic studies used to evaluate cholestatic jaundice involve a combination of clinical examination, liver function tests, imaging studies like ultrasound or magnetic resonance cholangiopancreatography (MRCP), and sometimes liver biopsy.[9][10][11][12]

Liver Chemistry Panel

Serum total and fractioned direct bilirubin should be obtained. A predominantly direct hyperbilirubinemia (more than 50% of total bilirubin) is observed in cholestasis. Serum alkaline phosphatase is often elevated 3 times greater than the upper standard limit of normal, while the hepatic transaminases may be normal or only mildly elevated. Serum albumin is usually normal except in cirrhosis and late chronic liver disease.

Hematology

Hematologic findings can assist in identifying the etiology of cholestatic jaundice, including:

- **Leukocytosis**: This finding may indicate cholangitis, alcohol-related hepatitis, and underlying malignancy.
- **Acute severe anemia**: Severe anemia may occur due to hemolysis. Evaluation of peripheral smear and reticulocyte count can help differentiate between these 2 entities. Chronic anemia can be seen in cirrhosis or underlying malignancy.
- **Elevated prothrombin time**: This finding can be seen with cholestasis, which rapidly reverses with vitamin K supplementation compared to patients with cirrhosis and hepatic synthetic dysfunction.

Radiological Evaluation

Abdominal ultrasound can help identify biliary ductal dilation and differentiate between hepatocellular causes of cholestasis (where the ducts are normal) and biliary obstruction (where the ducts are dilated). If dilated biliary ducts are encountered on initial ultrasound, MRCP can be used to assess the bile ducts further and identify stones, strictures, or features of malignancy. While a computed tomography scan can be helpful, MRCP has better sensitivity.

Liver Biopsy

A liver biopsy can be helpful in cases of suspected intrahepatic cholestasis when there is diagnostic uncertainty.

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The management of cholestatic jaundice depends upon the underlying etiology. The mainstay of obstructive cholestasis is biliary decompression.[13][14][15][16][17] Management of hepatocellular cholestasis includes treatment of the underlying disease and associated symptoms.

Biliary Decompression

Endoscopic sphincterotomy, with or without stent placement, can relieve obstruction due to bile duct stones. Similarly, dilation and stent placement can alleviate the obstruction in benign common bile duct strictures. Surgical resection of the obstructive lesion is preferred in the case of malignant obstruction and depends upon the disease stage and the patient's operative candidacy. Hepaticojejunostomy with Roux-en-Y bypass is an option if complete resection is not possible. If the patient is not a surgical candidate, palliation can be achieved with an endoscopically placed common bile duct stent, usually a metal stent. If endoscopic stent placement is not successful, a percutaneous transhepatic cholangiography tube can be placed for biliary decompression. Antibiotics are considered in the prebiliary and postbiliary decompression phases to reduce the risk of sepsis.

Management of Pruritus

In cases of obstructive physiology, pruritus may be relieved within 24 to 48 hours of biliary decompression. In other cases, medications may be needed for symptomatic control.[18] Medications options include:

- **Ursodeoxycholic acid**: This is recommended in doses of 13 to 15 mg/kg daily to treat the underlying condition in patients with cholestasis due to PBC and the pruritus associated with intrahepatic cholestasis of pregnancy.
- **Antihistamines**: These may be used for mild symptoms in patients with insomnia resulting from pruritus. They may benefit from the sedating effects.
- **Cholestyramine**: This bile acid sequestrant is considered first-line therapy for moderate to severe symptoms and is prescribed at a starting dose of 4 g to a maximum dose of 16 g, administered once or twice daily. Maximum dose is achieved by increasing the dose by 4 g daily every 4 weeks. Cholestyramine is best administered before breakfast to enhance the excretion of pruritogens, believed to have been stored in the gallbladder during the overnight fast. Patients should take the minimal effective dose to avoid adverse events, eg, nausea, steatorrhea, and fat-soluble vitamin malabsorption.
- **Rifampin**: This drug is used if cholestyramine does not relieve pruritus; this has been shown to improve pruritus when administered 150 mg to 300 mg twice daily. Serious adverse events include hepatitis, renal failure, and hemolytic anemia. Therefore, regular monitoring of aminotransferases is recommended.
- **Naltrexone**: This is an opioid antagonist administered at 12.5 mg to 50 mg daily. The efficacy is thought to be related to the role of endogenous opioids in the pathogenesis of cholestatic pruritus. This should not be prescribed to patients receiving opioid medications and should be used with caution in patients with underlying chronic pain conditions.
- **Sertraline**: This selective serotonin reuptake inhibitor has shown some efficacy in improving pruritus in small series when administered at 75 to 100 mg daily.

Nonpharmacological treatment options include ultraviolet light and plasmapheresis. These modalities are generally less effective, expensive, and not readily available for routine care of patients with pruritus. Surgical resection of less than 15% of the terminal ileum to prevent enterohepatic circulation (bile salt reabsorption) can be effective in intractable pruritus. This intervention may be reserved for refractory cases. Liver transplantation may be the only effective therapy for intractable cases of pruritus, but this is not a standard indication.

Tuberculosis

Treatment / Management

Latent Infection

TSTs and IGRAs cannot predict the progression from latent infection to active disease. Observational studies have demonstrated that the risk of developing active TB is highest during the first 2 years after acquiring infection and that reactivation is rare beyond 10 years after infection. The number needed to treat to prevent 1 case of active TB ranges from 36 in recently infected contacts to 179 in those remotely infected.[38] Treatment of latent TB reduces the likelihood of developing disease in populations at high risk. The evidence is less clear that treating patients at low or intermediate risk reduces the incidence.[34] According to the WHO, the rationale of treating patients with latent TB is based on the "probability that the condition will progress to active TB disease in specific risk groups, on the underlying epidemiology and burden of TB, the feasibility of the intervention, and the likelihood of a broader public health impact."[55]

The CDC recommends weekly INH plus rifapentine (3HP) for 3 months as the preferred preventive treatment regimen due to the short duration and decreased incidence of liver toxicity relative to the 6- and 9-month INH regimens (see **Image.** Recommendations for Regimens to Treat Latent TB infection). Reports of a flu-like syndrome occurring in recipients of the 3HP regimen initially raised concerns. However, a large, multinational trial found that while 11% of participants experienced symptoms of a systemic drug reaction (SDR), this is frequently time-limited within the first month of therapy. Of those experiencing SDR, 48% were able to complete treatment and serious adverse events were rare.[56]

The WHO guidelines on TB preventive therapy align with the CDC and include an alternative regimen of 1 month of daily rifapentine plus INH. The WHO also recommends that "in settings with high TB transmission, adults and adolescents living with HIV who have an unknown or a positive TST or IGRA and are unlikely to have active TB disease should receive at least 36 months of daily INH preventive therapy." The latter recommendation applies regardless of immune status and the administration of ART. Currently, no studies provide robust data to help guide TB preventive therapy for individuals having close contact with people infected with MDR-TB. The WHO recommends a targeted approach based on individual risk assessment.[55] Trials are progressing to ascertain the efficacy and safety of fluoroquinolones and delamanid. These trials will require long follow-ups to confirm the clinical endpoint of absence of disease. Devising TB preventive therapy protocols for various drug resistance patterns and assessing safety and efficacy in different populations is challenging (eg, children, adults, PLHIV, other immunosuppression, or comorbidities).[57][58][59] See StatPearls' companion resource, "Latent Tuberculosis," for more information on latent TB treatment.

Active Tuberculosis Infection

The goal of anti-TB therapy is to eradicate disease and eliminate transmission in all cases, an unambiguous goal that is exceedingly difficult to achieve in practice. Historically, anti-TB regimens have required many months of treatment, challenging patient adherence. These challenges exist everywhere and are particularly problematic in parts of the world with the highest TB prevalence, lacking public health infrastructure, and TB literacy. Straightforward and somewhat standardized TB treatment guidelines improve the ease of implementation globally; however, a standard "one-size-fits-all" 6-month regimen may be too long for some and not long enough for others. A pretreatment risk stratification algorithm could optimize and individualize treatment duration.[60][61]

The management of TB is very complex. Guidelines from the CDC and WHO provide detailed recommendations on dosing, first- and second-line anti-TB drugs, drug-drug interactions, management of treatment interruption, adverse events from treatment, culture-negative TB, extrapulmonary TB, HIV coinfection, children, advanced age, pregnancy, breastfeeding, and comorbidities (see **Image.** Drug Regimens for Microbiologically Confirmed Pulmonary Tuberculosis Caused by Drug-Susceptible Organisms).[WHO. TB Centers of Excellence for Training, Education, and Medical Consultation.]

Centers for Disease Control 2022 Interim Guidance

Standard anti-TB regimens cure most patients with drug-susceptible pulmonary TB, provided therapy can be completed. Until 2022, the CDC recommendations for drug-susceptible pulmonary TB disease had not changed in 50 years. The long-held standard regimen uses 4 drugs: INH, rifampin, ethambutol, and pyrazinamide. Before the development of this therapeutic combination, streptomycin was the first anti-TB drug used as monotherapy in the 1940s. Streptomycin initially provided significant benefit when administered for 6 months but ultimately failed due to the emergence of resistance.[62] The resistance to monotherapy, along with a better understanding of both the pathophysiology of tuberculosis and drug toxicities, informed decisions that led to the current standard of treatment.[63]

Current TB treatment has 2 phases. The initial intensive phase provides bactericidal activity to reduce the burden of rapidly replicating mycobacteria, reducing the chances of emerging resistance and decreasing lung inflammation, mortality, and transmission in most cases. The continuation phase sterilizes slowly replicating and dormant tubercle bacilli. In 2022, the CDC announced interim guidance on a 4-month anti-TB regimen for drug-susceptible pulmonary tuberculosis. The regimen consists of an intensive phase of 8 weeks of daily INH, rifapentine, pyrazinamide, and moxifloxacin, followed by a continuation phase of 9 weeks of daily rifapentine, INH, and moxifloxacin. This treatment option is available for patients older than 12, including PLHIV with CD4+ T lymphocyte counts greater than 100 cells/µL and taking an efavirenz-containing antiretroviral therapy regimen in the absence of any other known potential drug-drug interaction. Patients who are pregnant or breastfeeding, require treatment for extrapulmonary TB, use QT-prolonging medications, have a history of prolonged QT syndrome, or weigh less than 40 kg cannot take this regimen due to a lack of clinical trial data.[64][65]

Until 2022, the CDC standard pulmonary TB treatment had an intensive phase of therapy for 2 months with a 4-drug combination of INH, rifampin, ethambutol, and pyrazinamide. The continuation phase used a 2-drug combination of INH and rifampin for an additional 4 months. Drug susceptibility testing indicating the isolate's sensitivity to INH and rifampin allowed ethambutol discontinuation while maintaining INH, rifampin, and pyrazinamide for the remainder of the intensive phase. Patients undergoing anti-TB therapy require regular monitoring for clinical response and side effects. Evaluate sputum smears and cultures monthly until 2 consecutive cultures are negative. In patients with chest x-rays and evidence of cavitation who remain culture positive at 2 months, the recommendation is to extend the continuation phase for an additional 3 months (ie, a total of 9 months of therapy). People who are malnourished, active smokers, immunosuppressed, or PLHIV or extensive pulmonary disease may require an extended continuation phase.[63][62]

Drug-Resistant Tuberculosis

Compared to drug-susceptible TB treatment, drug-resistant TB poses a much greater risk of treatment failure and requires alternative and often prolonged regimens. Extended treatment regimens range from 18 to 21 months and incorporate a variety of first-line, second-line, new, and repurposed agents during the intensive and continuation phases of therapy. New and repurposed agents include but are not limited to delamanid, bedaquiline, pretomanid, linezolid, amoxicillin-clavulanate, meropenem-clavulanate, imipenem-cilastatin, cotrimoxazole, and macrolides. When prolonged regimens and injectable second-line agents are required, the risks of drug toxicities and patient difficulties with adherence are very high. Interventions to support adherence include psychological counseling and patient education, financial and material incentives, and mobile phone text reminders. New and repurposed agents may permit shorter durations of treatment in some circumstances. Adjuvant surgical excision may be necessary in efforts to eradicate cavities and non-viable lung tissue.[66] The WHO and American Thoracic Society/European Respiratory Society/Infectious Diseases Society of America detailed guidelines on managing drug-resistant TB. Clinicians should seek assistance from local health departments when caring for patients with suspected or confirmed drug-resistant TB.[13][67][68][69][70]

Bacille Calmette-Guerin Vaccine

BCG, a live attenuated strain of *M bovis*, has been a WHO-recommended vaccine for infants and children since 1974. This is the most widely used vaccine worldwide, preventing miliary TB and meningitis in infants in countries with high TB incidence. The vaccine is somewhat protective when administered to infants and children, but the protection wanes over several years. The vaccine does not provide significant protection when administered to adults. As previously discussed, the BCG vaccine can cause false-positive TSTs. The decision to initiate TB preventive drugs in patients who anticipate anti-tumor necrosis factor therapy or transplantation, have a possible history of the BCG vaccine, are IGRA-negative, and TST-positive is difficult. A person with a history of residence in a high TB prevalence region, no known prior TB infection, and a lack of radiological signs of latent or active TB could be at risk of developing reactivation TB once anti-TNF therapy or transplant commences. Discordant results, such as TST-positive/IGRA-negative, are common in people who received the BCG vaccine. No firm cutoff values are available for the size of TST induration in this particular setting. However, initiating TB preventive therapy is not without risks of adverse drug reactions.[71][72][73]

Unfortunately, this clinical situation cannot be addressed with absolute certainty. The guidelines approach this scenario somewhat indirectly and state the following: "Performing a second diagnostic test when the initial test is negative is one strategy to increase sensitivity. While this strategy to increase sensitivity may reduce the specificity of diagnostic testing, this may be an acceptable tradeoff in situations in which it is determined that the consequences of missing latent TB infection (ie, not treating individuals who may benefit from therapy) exceed the consequences of inappropriate therapy (ie, hepatotoxicity)." In this complicated situation, the consensus among most experts is to begin a regimen of TB preventive therapy several weeks before the start of the immunosuppressing intervention.[74]

Anemia

Treatment / Management

The treatment and management of acute anemia require prompt intervention to restore adequate oxygen delivery to tissues and prevent further complications. Treatment strategies often involve a combination of measures to stop ongoing bleeding, replace lost blood volume, and address the underlying cause of the anemia.

Initial Management for Acute Bleeding

The initial management of hemorrhage involves resuscitation and rapid blood volume replacement as clinically indicated until the patient is stabilized. Please see StatPearls' companion resource, "Hypovolemic Shock," for further information on the acute hemorrhage management. In general, this includes:

- Evaluating the ABCs
- Treating any life-threatening conditions immediately
- Administering supplemental oxygen
- Establishing 2 large-bore intravenous lines
- Intravenous fluid resuscitation (crystalloid is the initial fluid of choice)
- Applying direct pressure to any hemorrhage site, if possible

Acute Anemia Treatment

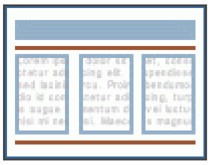
The primary treatment for acute anemia involves administering packed red blood cells (pRBCs) to replenish the lost blood volume. Typically, each unit of pRBCs is expected to increase the hematocrit by approximately 3 points and the hemoglobin level by roughly 1 point.

Transfusion thresholds

A restrictive transfusion strategy is generally followed for hospitalized, hemodynamically stable adult patients and those in critical care settings, with transfusion not recommended until the hemoglobin concentration drops to 7 g/dL or lower.[17] However, in patients with acute coronary syndrome, transfusion consideration arises when the hemoglobin level is ≤8 g/dL.[18] At very low hemoglobin concentrations (eg, 3 g/dL), the body's oxygen needs cannot be met, and tissue hypoxia occurs.[19]

For actively bleeding patients, transfusion decisions should be guided by clinical context and bleeding severity. A more liberal approach may be required to maintain hemodynamic stability until bleeding control is achieved during ongoing hemorrhage. In cases of massive hemorrhages, such as trauma or major surgical procedures, initiation of a massive transfusion protocol is warranted. This protocol involves promptly administering blood products, including pRBCs, fresh frozen plasma (FFP), platelets, and sometimes cryoprecipitate, to ensure hemodynamic stability and replenish coagulation factors.

All these blood components are integral to treating specific conditions associated with acute anemia. They may be considered for certain clinical indications (see **Table 2.** Treatment Options for Acute Anemia).



Table

Table 2. Treatment Options for Acute Anemia.

Sickle cell anemia

Treatment options for sickle cell anemia focus on symptom management, complication prevention, and enhancing overall quality of life. A blood transfusion may be initiated based on the rate of hemoglobin decline and the patient's clinical condition, especially during aplastic crises characterized by low reticulocyte counts. An exchange transfusion may be performed in vaso-occlusive crises or severe complications, eg, acute chest syndrome or stroke. This procedure involves gradually replacing the patient's blood with a donor or substitute, aiming to decrease sickle cell count, reduce blood viscosity, enhance circulation, and reduce further complications.

Hydroxyurea, an oral medication, is a viable option for managing sickle cell anemia. Its mechanism involves stimulating fetal hemoglobin production, thereby inhibiting the sickling of RBCs. Hydroxyurea effectively reduces the frequency and severity of sickle cell crises, decreases the need for transfusions, and improves overall symptoms and quality of life.

Platelet disorders

Patients with thrombocytopenia and clinical evidence of bleeding warrant a platelet transfusion. Those with platelet counts below 10,000/µL face a risk of spontaneous cerebral hemorrhage and thus necessitate prophylactic transfusion. For conditions like HUS and TTP, large-volume plasmapheresis with FFP replacement is the preferred treatment, often requiring daily sessions. Treatment objectives include increasing platelet count, decreasing LDH levels, and reducing RBC fragments as positive indicators of treatment response. Complementary measures, including corticosteroids and rituximab (a monoclonal antibody) are also often used to treat autoimmune TTP. Immunosuppressants such as cyclophosphamide and rarely splenectomy may be considered in chronic or relapsed TTP, especially when plasmapheresis and other treatments are not effective.

Initial management of atypical hemolytic uremic syndrome (aHUS) involves supportive care and plasmapheresis similar to the approach used for TTP. However, for patients with severe complement-mediated HUS, particularly those at risk of death or end-stage renal disease, eculizumab, a humanized monoclonal antibody to C5, is recommended. Emerging evidence suggests that early initiation of eculizumab can improve renal and nonrenal recovery.

The primary objective in treating immune thrombocytopenia (ITP) is to maintain a safe platelet count that mitigates clinically significant bleeding, rather than normalizing platelet counts. The bleeding risk occurs when the platelet counts fall below 30,000/µL. Immediate platelet transfusion is recommended for patients experiencing severe bleeding, eg, intracranial or GI bleeding, and having a platelet count of <10,000/µL. In addition to platelet transfusion, specific therapies for ITP, including intravenous immune globulin, glucocorticoids, rituximab and romiplostim, are commonly used.

Congenital bleeding disorders

Von Willebrand disease, characterized by deficient or defective von Willebrand factor, can be effectively managed using different approaches. Primary treatment options include desmopressin, recombinant von Willebrand factor, and von Willebrand factor/factor VIII concentrates. These treatments aim to replenish or enhance the function of von Willebrand factor, thereby mitigating bleeding episodes and improving overall clinical outcomes.

Factor VIII and IX concentrates are employed to manage hemophilia A (factor VIII deficiency) and hemophilia B (factor IX deficiency), respectively. The dosage and administration of these concentrates vary depending on the site and severity of bleeding in each patient. This tailored approach ensures effective control of bleeding episodes while minimizing the risk of complications associated with hemophilia.

Disseminated intravascular coagulation

The treatment of DIC is focused on addressing the underlying cause and managing the associated bleeding and thrombotic complications. The primary step is treating the underlying condition, eg, infection, trauma, malignancy, or obstetric complications. Supportive care is essential and may involve transfusion of FFP to replace depleted clotting factors, and platelet transfusions to correct thrombocytopenia. In cases of severe bleeding or organ failure, cryoprecipitate may be used to replenish fibrinogen. For thrombotic manifestations, heparin may be administered cautiously in selected cases, especially in the presence of microthrombi and organ ischemia. Antifibrinolytic agents (eg, tranexamic acid) are contraindicated in DIC due to the risk of exacerbating bleeding.

Hypoglycemia

Treatment / Management

Identification of a hypoglycemic patient is critical due to potential adverse effects, including coma or death. Severe hypoglycemia can be treated with intravenous (IV) dextrose followed by an infusion of glucose. For conscious patients able to take oral (PO) medications, readily absorbable carbohydrate sources (such as fruit juice) should be given. For patients unable to take oral agents, glucagon should be administered. Glucagon can be given intramuscularly or intranasally using the newest available formulations.[9][10] Once the patient is awake, a complex carbohydrate food source should be given to achieve sustained euglycemia. More frequent blood glucose monitoring should occur to rule out further drops in blood sugar.

Nonpharmacological management of recurrent hypoglycemia involves patient education and lifestyle changes. Some patients are unaware of the serious ramifications of persistent hypoglycemia. As such, patients should be educated on the importance of routine blood glucose monitoring and on identifying the individual's symptoms of hypoglycemia. Pharmacologic intervention should be modified if lifestyle changes are ineffective in preventing further episodes. Patients should be advised to wear a medical alert bracelet or necklace and carry a glucose source like gel, candy, or tablets in their purse in case symptoms arise. In the outpatient setting, reviewing blood sugar and food logs may help identify problem areas for the patient.

Glycemic control has been an important aspect of medical management due to the association between glycated hemoglobin levels and cardiovascular events in diabetes mellitus type 2 patients. In the 2008 ACCORD trial, it was determined that intensive therapy (defined as a goal hemoglobin A1C less than 6.0%) did not significantly reduce major cardiovascular events and was associated with increased mortality and risk for hypoglycemia.[11] However, it should be noted that the intensive therapy group had proportionally more participants using rosiglitazone compared to the standard therapy group (91.2% versus 57.5%), thus possibly contributing to an increased incidence of cardiovascular events in the intensive therapy group.

The 2009 VADT study additionally studied the effect of intensive blood glucose control in a sample of 1791 veterans with poorly controlled diabetes mellitus type 2. More rigid glycemic control did not appear to significantly affect cardiovascular outcomes, although it did improve microalbuminuria compared to the standard therapy arm.[12] However, the results cannot be extrapolated to females since 97% of the study participants were male. Besides, there was a significant dropout (approximately 15%), limiting statistical power.

Regarding endogenous sources of insulin, insulinomas are often managed surgically. Evidence of an insulinoma should prompt workup or investigative efforts into potential multiple endocrine neoplasia (MEN) disorders.

Thalassemia

Treatment / Management

Thalassemia treatment depends on the type and severity of the disease.

Mild thalassemia (Hb: 6 to 10g/dl):

Signs and symptoms are generally mild with **thalassemia** minor, and little, if any, treatment is needed. Occasionally, patients may need a blood transfusion, particularly after surgery, following childbirth, or to help manage **thalassemia** complications.

Moderate to severe thalassemia (Hb less than 5 to 6g/dl):

- **Frequent blood transfusions:** More severe forms of **thalassemia** often require regular blood transfusions, possibly every few weeks. The goal is to maintain hemoglobin (Hb) at approximately 9-10 mg/dL to promote patient well-being and to monitor erythropoiesis and suppress extramedullary hematopoiesis. To limit transfusion-related complications, washed, packed red blood cells (RBCs) at approximately 8 to 15 mL cells per kilogram (kg) of body weight over 1 to 2 hours are recommended.
- **Chelation therapy:** Due to chronic transfusions, iron begins to accumulate in various organs of the body. Iron chelators (deferasirox, deferoxamine, deferiprone) are given concomitantly to remove extra iron from the body.
- **Stem cell transplant:** Stem cell transplant, (bone marrow transplant), is a potential option in selected cases, such as children born with severe **thalassemia**. It can eliminate the need for lifelong blood transfusions.[8] However, this procedure has complications, and the clinician must weigh them against the benefits. Risks include graft-versus-host disease, chronic immunosuppressive therapy, graft failure, and transplantation-related mortality.[9]

- **Gene therapy** is a recent advance in the management of severe **thalassemia**. It involves harvesting the patient's autologous hematopoietic stem cells (HSCs) and genetically modifying them with vectors that express the normal genes. These are then reinfused into patients after they have undergone the required conditioning to eliminate existing HSCs. The genetically modified HSCs produce normal hemoglobin chains, and normal erythropoiesis ensues.
- **Genome editing techniques:** Another recent approach is the editing of genomic libraries using zinc-finger nucleases, transcription activator-like effectors, and clustered regularly interspaced short palindromic repeats (CRISPR) with the Cas9 nuclease system. These techniques target specific mutation sites and replace them with the normal sequence. The limitation of this technique is its inability to produce a sufficient number of corrected genes to cure the disease.[10]
- **Splenectomy:** Patients with **thalassemia** major often undergo splenectomy to limit the number of required transfusions. Splenectomy is the usual recommendation when the annual transfusion requirement exceeds 200-220 mL RBCs/kg/year, with a hematocrit of 70%. Splenectomy not only limits the number of required transfusions but also controls the spread of extramedullary hematopoiesis. Postsplenectomy immunizations are necessary to prevent bacterial infections, including *Pneumococcus*, *Meningococcus*, and *Haemophilus influenzae*. Postsplenectomy sepsis is possible in children, so this procedure is deferred until 6 to 7 years of age, and then penicillin is given for prophylaxis until they reach a certain age.
- **Cholecystectomy**: Patients may develop cholelithiasis due to increased hemoglobin (Hb) breakdown and bilirubin deposition in the gallbladder. If it becomes symptomatic, patients should undergo cholecystectomy at the same time as they undergo splenectomy.

Diet and exercise:

Reports indicate that drinking tea reduces iron absorption from the gastrointestinal tract. In patients with **thalassemia**, tea may be a healthy beverage to consume routinely. Vitamin C helps in iron excretion from the gut, especially when used with deferoxamine. But using vitamin C in large quantities and without concomitant deferoxamine use, there is a higher risk for fatal arrhythmias. Therefore, the recommendation is to use low doses of vitamin C in combination with iron chelators (e.g., deferoxamine).[10]

Lymphoma

Treatment / Management

Once a gastric MALT **lymphoma** diagnosis is made, the patient should be treated with *H pylori* eradication therapy according to current guidelines, considering prior antibiotic exposure, allergies, and local antibiotic resistance patterns. Noninvasive testing should be performed to confirm eradication of *H pylori* using either a urea breath test or stool antigen test 6 weeks after the completion of the therapy. Notably, even if patients are confirmed *H pylori*–negative with 2 different forms of testing before eradication therapy, they should still undergo a 2-week course of antibiotics and proton pump inhibitors per *H pylori* treatment guidelines. Prior exposure to macrolide therapy imparts resistance to *H pylori*, potentially reducing eradication rates if reused.[28][29] Gastric MALT lymphomas have been shown to respond to *H pylori* eradication therapy in more than 75% of cases, regardless of bacterial presence at the time of **lymphoma** diagnosis. One possible theory is that infection with other unidentified bacteria may cause MALT pathology.[30]

In patients with early-stage low-grade gastric MALT lymphomas (modified Lugano stages I-IIE), an EGD should be performed 3 to 6 months after *H pylori* eradication. During this endoscopy, gastric mapping should be performed with a histological evaluation of each specimen using the Sydney protocol. Complete remission or histological response is determined if there is a total resolution of diffuse lymphoid infiltrates and lymphoepithelial lesions in biopsy specimens on 2 consecutive endoscopies. Notably, evidence of histological remission may take 1 to 14 months post–*H pylori* eradication to become apparent.[31][32] If an experienced pathologist determines complete remission, complete histological response, or probable minimal residual disease, the patient should undergo EGD with biopsies every 6 months for the first 2 years. After 2 years, the exam frequency can be extended to EGD with gastric mapping biopsies every 12 to 18 months.

Patients with *H pylori*–negative gastric MALT **lymphoma** who have limited-stage (I-II_1) disease should be managed with involved-site radiation therapy (or ISRT) or single-agent rituximab in patients with contraindications to radiotherapy.[33] When patients show evidence of incomplete eradication or residual disease of low-grade gastric MALT **lymphoma** after *H pylori* treatment, they should undergo an EGD with gastric mapping biopsies every 3 to 6 months, following an expectant observation approach. If patients show no change in low-grade gastric MALT **lymphoma** after *H pylori* eradication therapy, they can opt for EGDs every 3 to 6 months with gastric mapping biopsies for expectant observation, or they can choose treatment with localized radiation or chemotherapy, with or without rituximab.

Patients with higher-grade gastric MALT **lymphoma** (Lugano stages II2-IV) are generally observed expectantly with clinical surveillance alone. If any indications for treatment arise, such as gastrointestinal bleeding, constitutional symptoms, or bulky disease, chemoimmunotherapy is preferred. The most commonly utilized regimens are based on a rituximab + chemotherapy backbone, such as R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone) or BR (bendamustine-rituximab).[33][34] Overt histological transformation to DLBCL should be treated empirically as DLBCL. Please see StatPearls' companion resource, "Diffuse Large B-cell **lymphoma**," for further information.

Progressive gastric MALT (non-DLBCL) **lymphoma** after first-line rituximab-based chemotherapy can be treated with various salvage regimens with either rituximab + lenalidomide or oral BTK or PI3K-delta inhibitors. Clinical trial enrollment should always be considered in the relapsed or refractory setting.[33] Management of salvage therapy and clinical trials should always be guided by experienced hematologists at high-volume centers.

Surgery should be reserved for patients with acute bleeding not amenable to endoscopic therapy, perforation, or obstruction. Clinicians should inform patients that remission rates with radiation are higher than with chemotherapy or immunotherapy in gastric MALT **lymphoma**, with some studies reporting more than 95% complete histologic remission rates. Immunotherapy with rituximab increases the risk of hepatitis B reactivation and should be discussed with patients before choosing treatment options. Screening for HIV and hepatitis B and C should be completed before starting rituximab.

Sickle Cell Crisis

Treatment / Management

Treatment and management of SCD primarily consists of managing symptoms and associated complications. Curative therapies such as stem **cell** transplant and gene editing are new, more complex, and costly. Unfortunately, nations rife with **sickle cell** disease may not have healthcare systems able to initiate these interventions.[29]

Vaso-Occlusive Crisis Management

As a general rule, treatment requires an early diagnosis, prevention of complications, and management of end-organ damage.[6] Rapid pain assessment and initiation of analgesia should be undertaken promptly. Depending on the degree and severity of the pain, analgesic administration can be given intravenously (IV) or intranasally. Oral analgesics can be used for patients who are not in severe pain and can tolerate oral medications. Generally, the type, route, and dose of the analgesic should be individualized to the patient. Most guidelines recommend early initiation of parenteral opioid analgesics, usually with morphine at 0.1 mg/kg IV or subcutaneously (SC) every 20 minutes, and maintaining this analgesia with morphine at doses of 0.05 to 0.1 mg/ kg every 2 to 4 hours (SC/IV or orally). Those with persistent pain benefit from a PCA pump. Close monitoring of vital signs, including oxygen saturation, should be maintained with frequent pain severity or resolution reassessments.[28] If the pain is controlled, the patient may be ready for discharge with a home care plan and oral analgesia.[30] If the pain is uncontrolled despite the above treatment plan, consider hospitalization and the use of more potent forms of analgesia or higher doses titrated to the patient's needs. Simple or exchange transfusion may be warranted.[31]

Maintaining adequate hydration and being vigilant in identifying other causes of pain that may need additional treatment is prudent. Dehydration is one of the known factors of venous-occlusive disease, caused by the formation of deoxygenated HbS (deoxyHbs), which causes increased adherence and decreased flexibility of the erythrocytes. Plasma concentration is a function of the intracellular content of deoxy HbS. The sickling of erythrocytes depends on both factors. This is why patients who present with a painful **crisis** should be given fluids to slow down or stop the sickling process. Increasing the plasma volume decreases blood viscosity and indirectly reduces the amount and concentration of HbS within the red blood cells.

For other entities like acute chest syndrome, splenic sequestration-supportive care with oxygen, judicious fluid administration, and transfusion therapy are needed. Close monitoring of oxygen saturation and respiratory status is also necessary, with particular attention to excessive sedation.[32] Empiric antibiotics, adequate analgesics, and simple or exchange transfusion may be considered for acute chest syndrome. Incentive spirometry, oral hydration, and comfort measures are recommended. Patients with splenic sequestration **crisis** resulting in hypovolemic shock, if not treated aggressively, have higher mortality. Management requires aggressive supportive care and blood transfusion.[28]

Hydrea is an antimetabolite, a chemotherapy agent, that is foremost in treating **sickle cell** disease by augmenting HbF levels within the red blood cells. This mechanism decreases the rate of acute chest syndrome, pain, and other sequelae.[33] As a result, the quality of life is improved, hospitalizations are lessened, and transfusion needs are reduced.[34] Glutamine is required to synthesize amino acids that protect the erythrocyte against oxidative damage.[6] It does so by increasing the availability of Nicotinamide Adenine Dinucleotide (NAD), a redox factor.[5] NAD does not affect the Hgb, Hct, or reticulocyte count. Experts believe NAT reinstates the intestinal barrier, prohibiting bacterial entry.[35] These microbial interlopers are believed to compound the inflammation that promotes **vaso-occlusive crises**. Vaxelotor modifies HbS so that its polymerization is obstructed and its oxygen affinity increases.[6][5] It improves hemolysis but has little effect on the **sickle crisis**. Vaxelotor effects with glutamine are synergistic; its effects with crizanlizumab are antagonistic. As an oxygen affinity modulator, vaxelotor reportedly has a relatively good safety profile and is used foremost in Europe.[36] Crizanlizumab is a humanized monoclonal antibody gG-2 Kappa that blocks P-selectin, the binding protein of **sickle** cells.[5][6] By doing so, crizanlizumab interrupts the **vaso-occlusive** process. It additionally blocks the adhesion of neutrophils and activated platelets.[36] Crizanlizumab is administered by monthly infusion.

Bone marrow transplant (BMT) has curative potential in **sickle cell** disease.[6] Hematologic stem **cell** transplant (HSCT) with HLA-matched sibling donors and either myeloablative or reduced-intensity conditioning regimens have a 5-year overall survival (OS) of about 90%. Matched unrelated transplants have difficulties with GVHD as high as 29%. Haploidentical HSCT, with posttransplant cyclophosphamide, has some utility. Umbilical cord transplant has shortcomings in terms of a low transplant **cell** dose and, if from an unmatched donor, the increased risk of graft rejection and GVHD. Gene therapy, with autologous transplantation, is developing into a possible modality for **sickle cell** treatment.[37] Functional human beta-globin genes (eg, Lentiglobin) are inserted ex-vivo into a patient's stem cells which are then transplanted back into the patient. These patients then produce HbA and have noteworthy reductions in hemolysis and **vaso-occlusive crises**.

Long-term management of **sickle cell** disease is focused on the prevention of infection as well as the prevention of end-organ damage.[13] Besides lifelong administration of penicillin V, vaccinations against meningococcus, pneumococcus, and hemophilus influenza B are available to prevent infection with encapsulated organisms. COVID-19 and influenza vaccinations should be encouraged. Folate supplementation helps prevent a functional folate deficit.